

STM32CubeIDE Setup Guide for MRSUBG Tx/Rx Wake-Up Radio Project

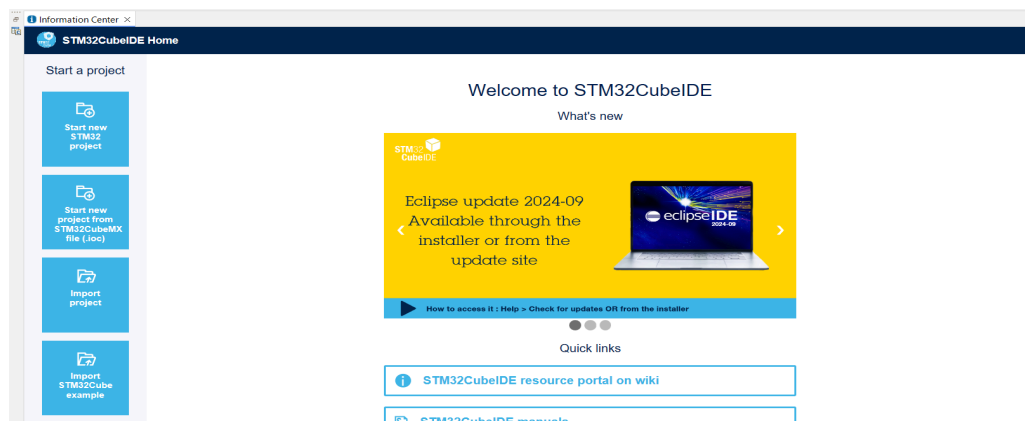
 STM32CubeIDE Version: 1.19.0

 Project: Wake-Up Radio (LPAWUR) with MRSUBG_BasicGeneric_Tx/Rx Example

Step-by-Step Instructions

1. Launch STM32CubeIDE

- After opening the software, **select or create a new workspace**.



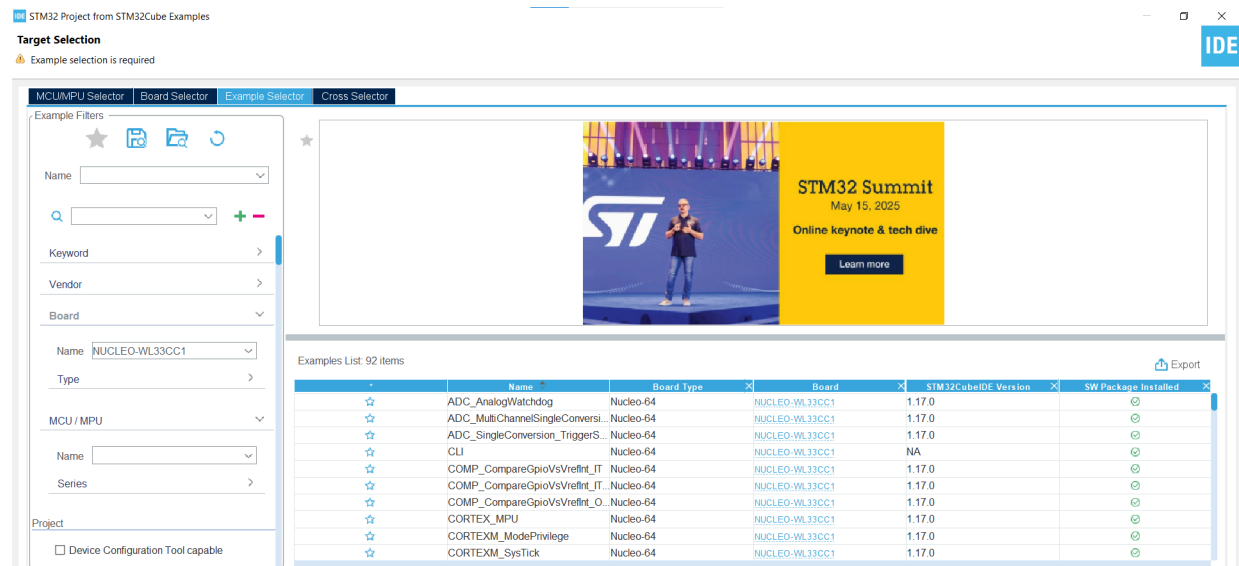
2. Import the STM32Cube Example

- At the bottom-left corner, click on **“Import STM32Cube Example”**.



3. Select Your Board

- In the search bar under “**Board**”, enter your board name (e.g., **NUCLEO-WL33CC1**).



4. Choose Example Project

- A list of available examples will appear.

Examples List: 92 items

Name	Board Type	Board	STM32CubeIDE Version	SW Package Installed
LPAWUR_Skeleton	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
LPAWUR_WakeupRadio_Rx	Nucleo-64	NUCLEO-WL33CC1	NA	✓
LPUART_TwoBoards_CoMT	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
LPUART_WakeUpFromDeepStop	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
MRSUBG_802_15_4_Rx	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
MRSUBG_802_15_4_Tx	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
MRSUBG_BasicGeneric_Rx	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
MRSUBG_BasicGeneric_Tx	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
MRSUBG_Chat	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓
MRSUBG_DatabufferHandler_Rx	Nucleo-64	NUCLEO-WL33CC1	1.17.0	✓

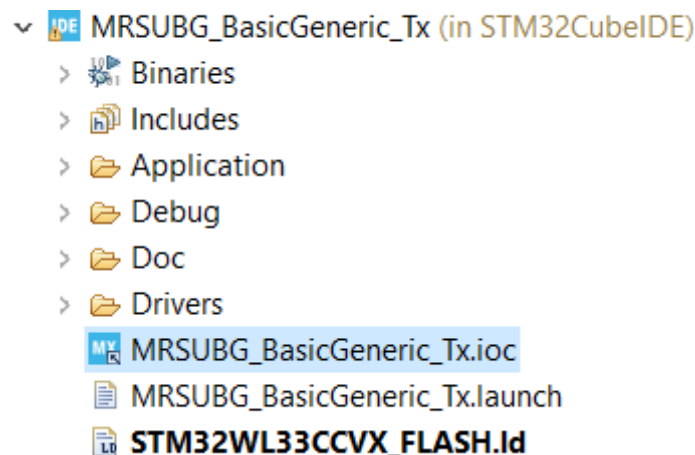
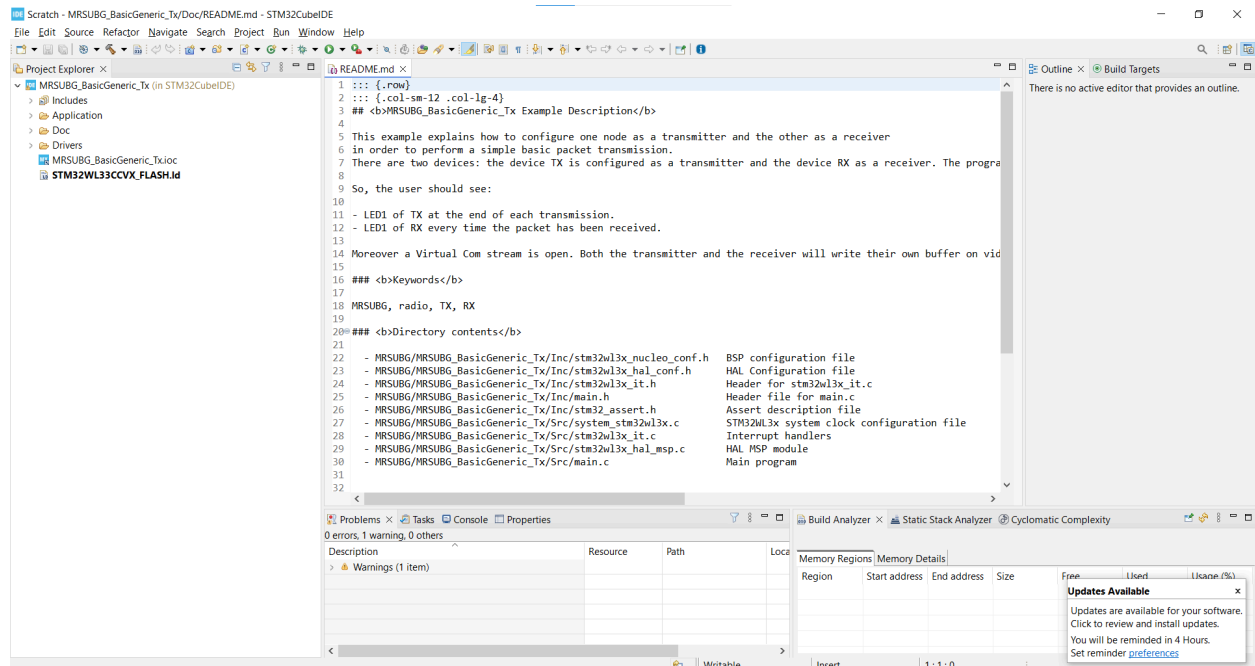
- Search for: **MRSUBG_BasicGeneric_Tx** or **MRSUBG_BasicGeneric_Rx**

MRSUBG_BasicGeneric_Rx	Nucleo-64	NUCLEO-WL33CC1	1.17.0
MRSUBG_BasicGeneric_Tx	Nucleo-64	NUCLEO-WL33CC1	1.17.0

- Click on your chosen project and then click **Next > Finish**.

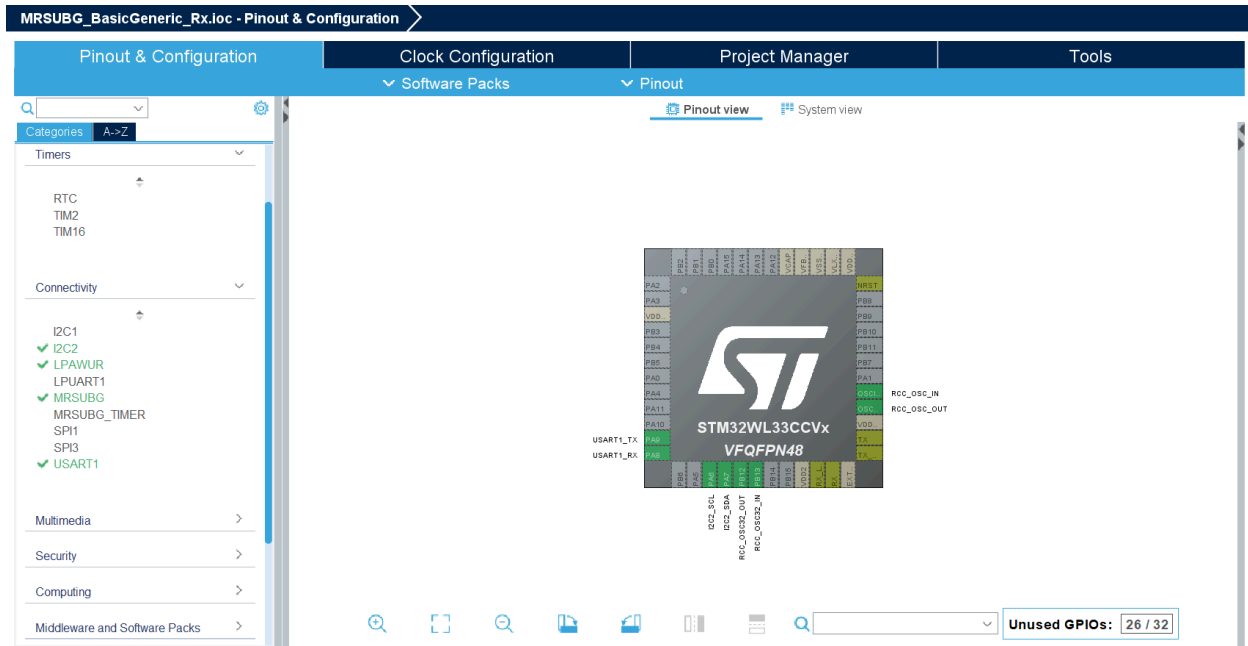
5. Access the Project IOC Configuration

- Once the project is open, locate and double-click the **.ioc** file.

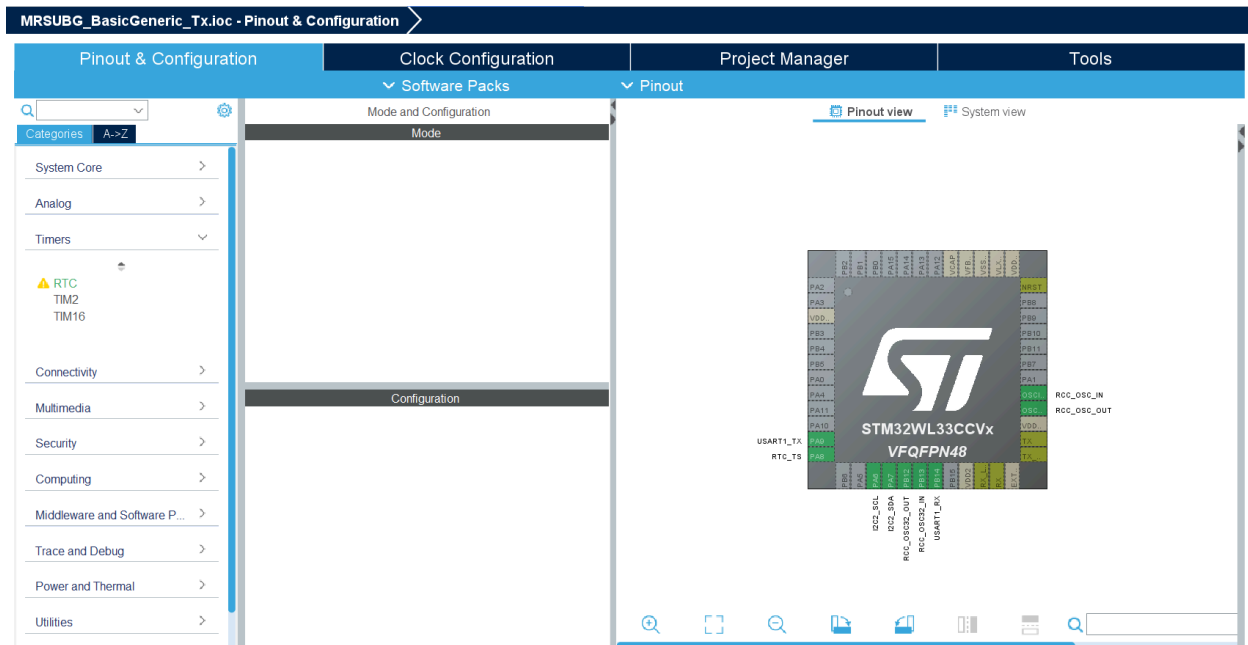


6. Enable All the System's Features

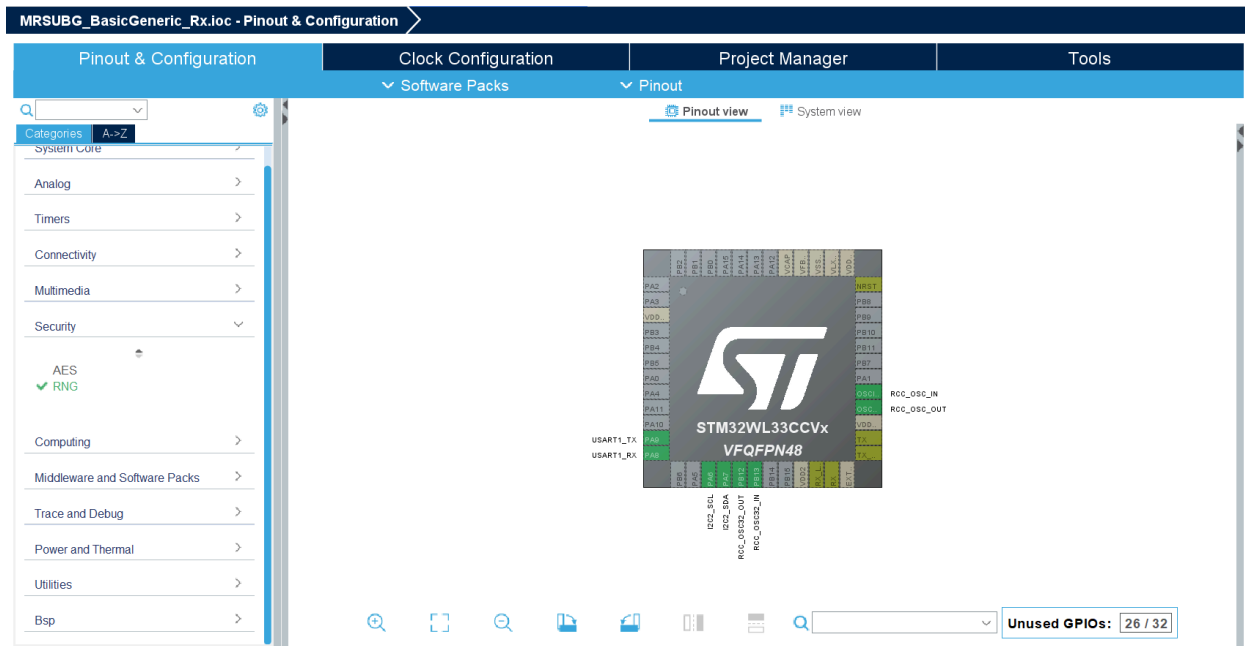
- In the IOC editor:
 - Go to **Connectivity**, and activate **LPAWUR** (Low Power Wake-Up Radio) and **I2C2** to enable sensors.



- Next, go to **Timers** and enable **RTC**, this is crucial for Waking Up for both main node and mesh node from sleep.



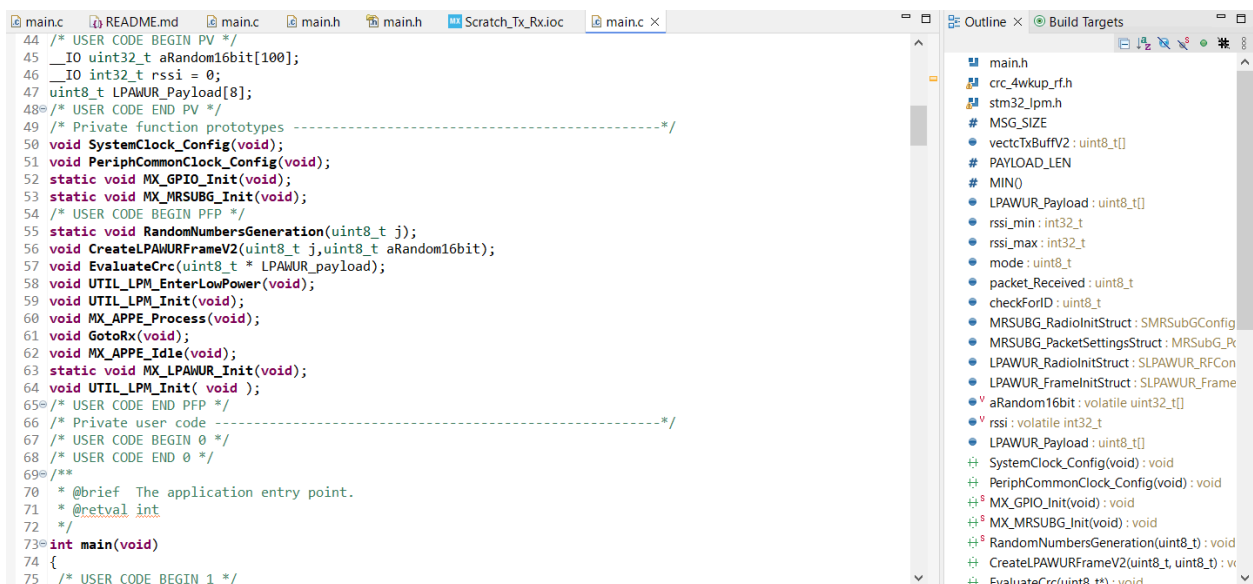
- Afterwards, go to **Security** and enable **RNG**, this is used to randomize packet transmission timing.



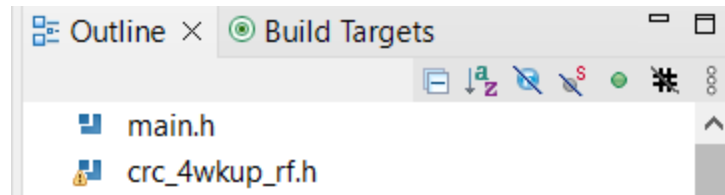
- Save with **Ctrl + S**.

7. Add Your Source Code

- Go to **Application/User/main.c** and paste the code.



- To find **main.h** double click on the file on the top-left corner and paste the code.



8. Copy Additional Folders (Drivers,LPM, WUR, TEMPHUMID & SCHEDULER)

- These folders contain essential drivers for the wake-up functionality.
- For drivers, normally at this path :
C:\Users\mzakri\STM32CubeIDE\Scratch\MRSUBG_BasicGeneric_Rx\STM32CubeIDE\Drivers\STM32WL3x_HAL_Driver :
- Copy and paste all the drivers into this path.

mzakri > STM32CubeIDE > Scratch > MRSUBG_BasicGeneric_Rx > STM32CubeIDE > Drivers > STM32WL3x_HAL_Driver				
Nom	Modifié le	Type	Taille	
stm32wl3x_hal	01/04/2025 11:32	Fichier source C	16 Ko	
stm32wl3x_hal	01/04/2025 11:32	Fichier source C H...	12 Ko	
stm32wl3x_hal_cortex	01/04/2025 11:32	Fichier source C	13 Ko	
stm32wl3x_hal_cortex	01/04/2025 11:32	Fichier source C H...	16 Ko	
stm32wl3x_hal_def	01/04/2025 11:32	Fichier source C H...	7 Ko	
stm32wl3x_hal_dma	01/04/2025 11:32	Fichier source C	27 Ko	
stm32wl3x_hal_dma	01/04/2025 11:32	Fichier source C H...	28 Ko	
stm32wl3x_hal_flash	01/04/2025 11:32	Fichier source C	16 Ko	
stm32wl3x_hal_flash	01/04/2025 11:32	Fichier source C H...	22 Ko	
stm32wl3x_hal_flash_ex	01/04/2025 11:32	Fichier source C	12 Ko	
stm32wl3x_hal_flash_ex	01/04/2025 11:32	Fichier source C H...	3 Ko	
stm32wl3x_hal_gpio	01/04/2025 11:32	Fichier source C	21 Ko	
stm32wl3x_hal_gpio	01/04/2025 11:32	Fichier source C H...	17 Ko	
stm32wl3x_hal_gpio_ex	01/04/2025 11:32	Fichier source C H...	6 Ko	
stm32wl3x_hal_i2c	01/04/2025 11:32	Fichier source C	241 Ko	
stm32wl3x_hal_i2c	01/04/2025 11:32	Fichier source C H...	39 Ko	
stm32wl3x_hal_i2c_ex	01/04/2025 11:32	Fichier source C	8 Ko	
stm32wl3x_hal_i2c_ex	01/04/2025 11:32	Fichier source C H...	6 Ko	
stm32wl3x_hal_lpawur	01/04/2025 11:32	Fichier source C	11 Ko	
stm32wl3x_hal_lpawur	01/04/2025 11:32	Fichier source C H...	9 Ko	

- Navigate to your workspace folder (e.g.,

C:\Users\<YourName>\STM32CubeIDE\Scratch\MRSUBG_BasicGeneric_Tx\STM32CubeIDE\Application\User).

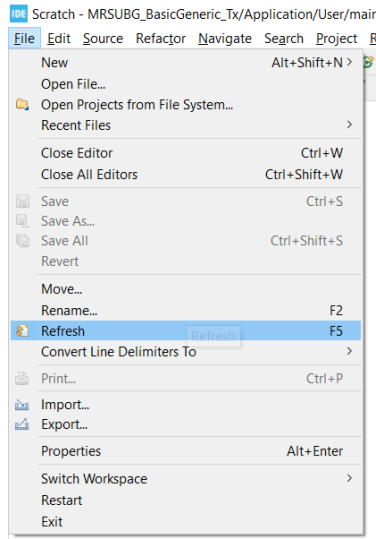
Utilisateurs > mzakri > STM32CubeIDE > Scratch > MRSUBG_BasicGeneric_Tx > STM32CubeIDE > Application > User				
Nom	Modifié le	Type	Taille	
 syscalls	01/04/2025 11:32	Fichier source C	3 Ko	
 sysmem	01/04/2025 11:32	Fichier source C	3 Ko	

- Copy the **LPM**, **WUR**, **TEMPHUMID** & **SCHEDULER** folders into your project directory under the **User** folder :

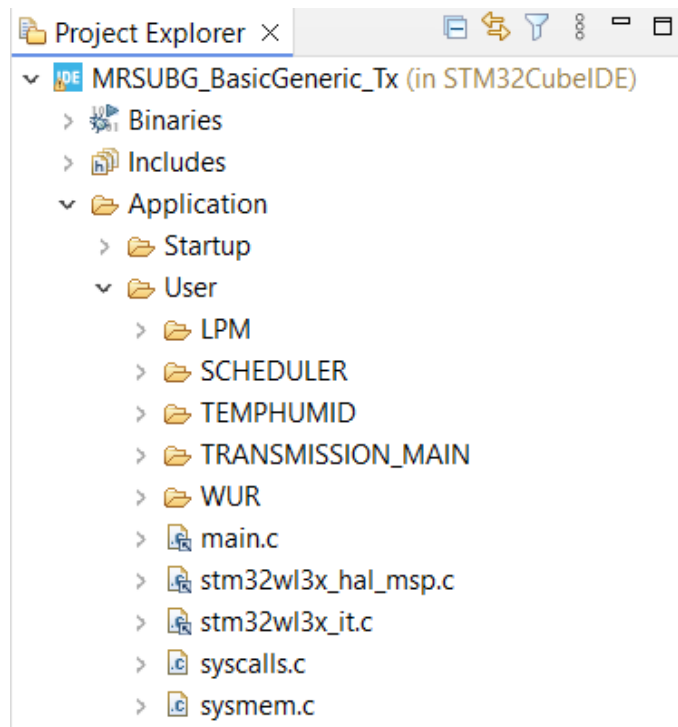
Utilisateurs > mzakri > STM32CubeIDE > Scratch > MRSUBG_BasicGeneric_Tx > STM32CubeIDE > Application > User >				
Nom	Modifié le	Type	Taille	
 LPM	11/06/2025 10:05	Dossier de fichiers		
 SCHEDULER	11/06/2025 10:05	Dossier de fichiers		
 TEMPHUMID	11/06/2025 10:05	Dossier de fichiers		
 TRANSMISSION_MAIN	11/06/2025 14:40	Dossier de fichiers		
 WUR	11/06/2025 10:05	Dossier de fichiers		
 syscalls	01/04/2025 11:32	Fichier source C	3 Ko	
 sysmem	01/04/2025 11:32	Fichier source C	3 Ko	

9. Refresh the IDE

- After copying, close the file explorer and return to STM32CubeIDE.
- **File > Refresh.**



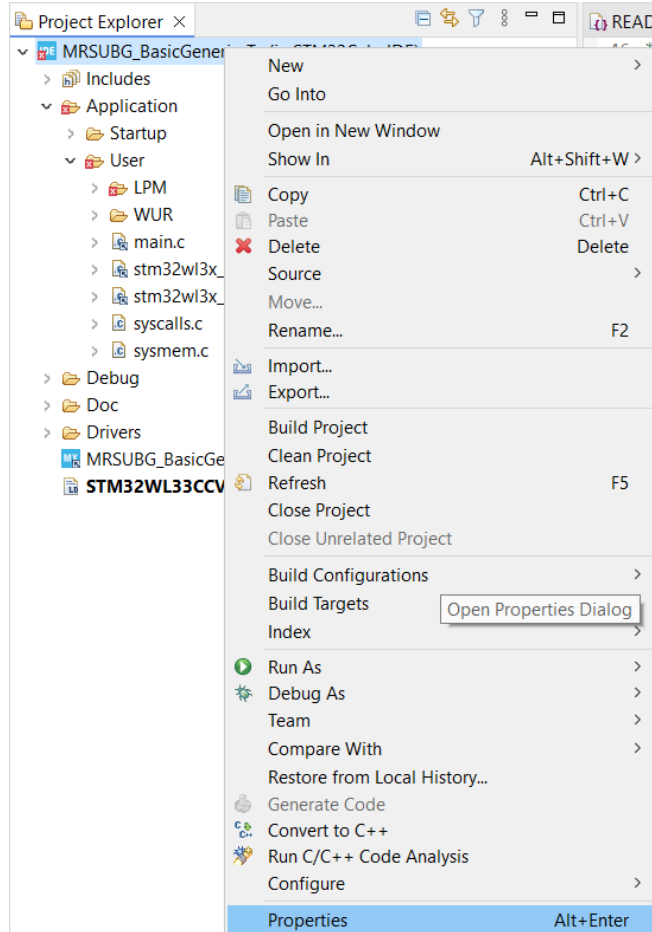
- The folders should now appear in the Project Explorer.



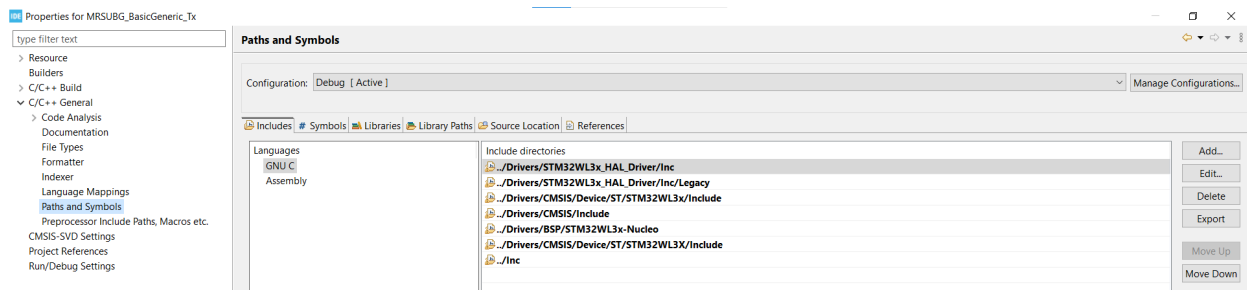
10. Add Include Paths

 Without this step, the compiler won't recognize the new folders.

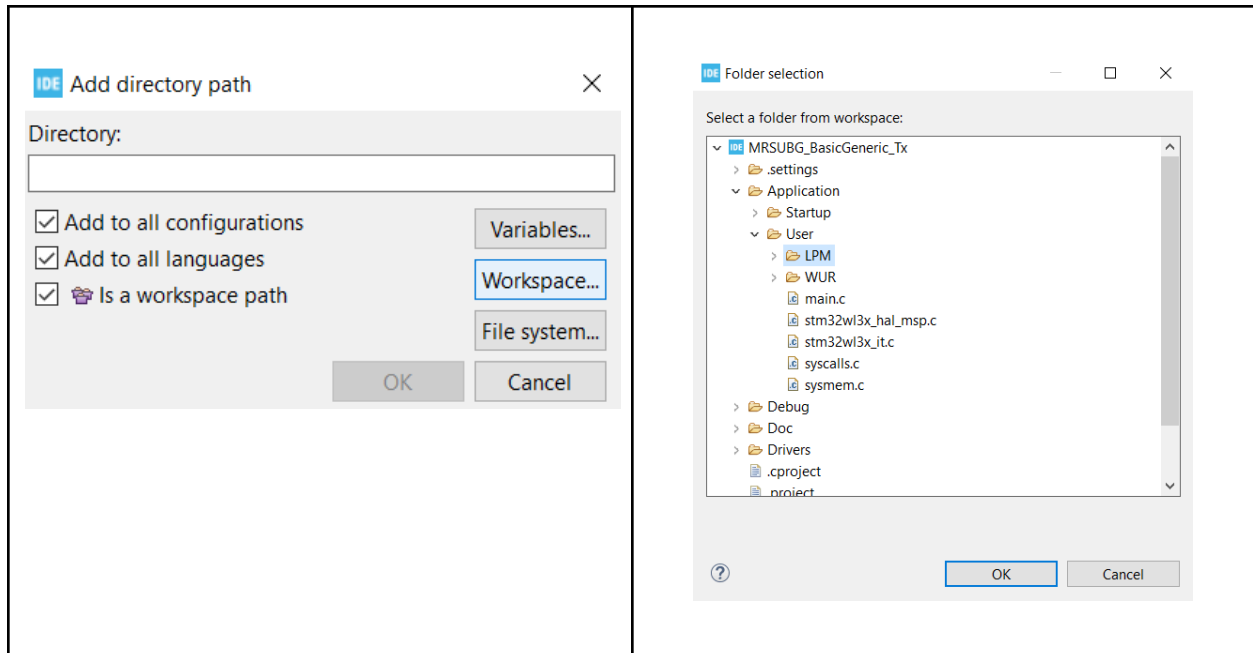
- **Right-click the project > Properties.**



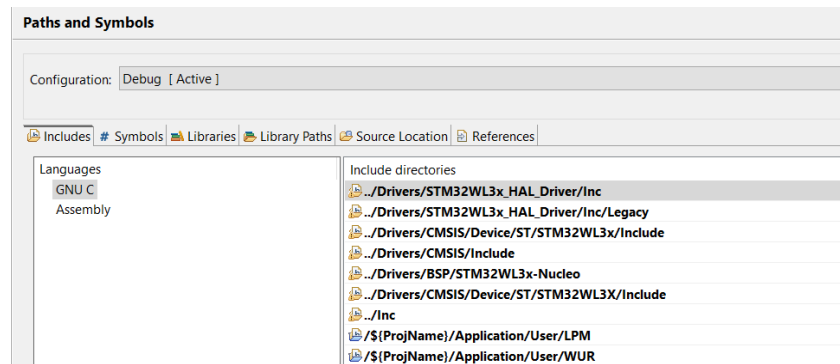
- Go to **C/C++ General > Paths and Symbols**.



- Under the **Includes** tab, click **Add...** > **Workspace** > select **LPM**, check all boxes > **OK**.



- Repeat for the **WUR** folder.



- Apply and close the settings, then **refresh the IDE again**.

11. Build the Project

- Click on the **hammer icon**  to build.
- Warnings may appear the first time but **should disappear on the second build**.

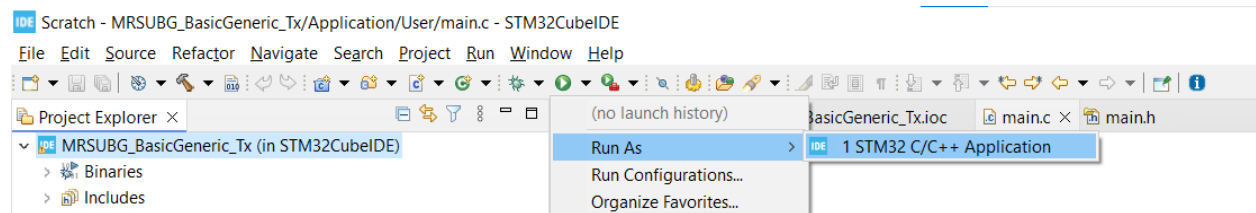
text	data	bss	dec	hex	filename
31336	156	5736	37228	916c	MRSUBG_BasicGeneric_Tx.elf

Finished building: default.size.stdout

10:48:05 Build Finished. 0 errors, 0 warnings. (took 330ms)

12. Run the Program

- Click the **green triangle icon** to start debugging.
- Then click **STM32 C/C++ Application** then **OK** to run.



13. Select Main or Mesh Node

- Open `main.c` and make sure the initialization has this part in `"int main(void)"` :
 - Main Node

```
printf("STM32WL3 LPAWUR - Main Node Started\n\r");
```

- Mesh Node

```
printf("STM32WL3 LPAWUR - Mesh Node.\n\r");
```

 Make sure both transmitter and receiver boards are powered and flashed accordingly.

Troubleshooting Tips

Issue	Solution
Board not showing up	Ensure your USB is connected, try restarting STM32CubeIDE

Project fails to build	Make sure LPM/WUR paths are included, and files are properly copied
Warnings during build	Ignore if they appear on first build — recompile
No output observed	Confirm Tx and Rx boards are flashed with corresponding roles
Not compiling	If you wanted to change something in the .ioc file, there's a high chance of the program not working anymore -> delete the whole project and redoing everything from start. NO KNOWN FIX YET FOR THIS BUG.

You're Done!

You now have a working setup for low-power wake-up radio communication using STM32CubeIDE and the STM32WL Nucleo boards.